

07

/ 13 LECTURES

Routing and Forwarding — Part I

Routers, routing tables, Longest Prefix Match, static routes, default routes, hop-by-hop forwarding, and why TTL exists.

Router Anatomy

Routing Table

Longest Prefix Match

Static Routing

Default Route 0.0.0.0/0

Packet Forwarding

Hop-by-Hop

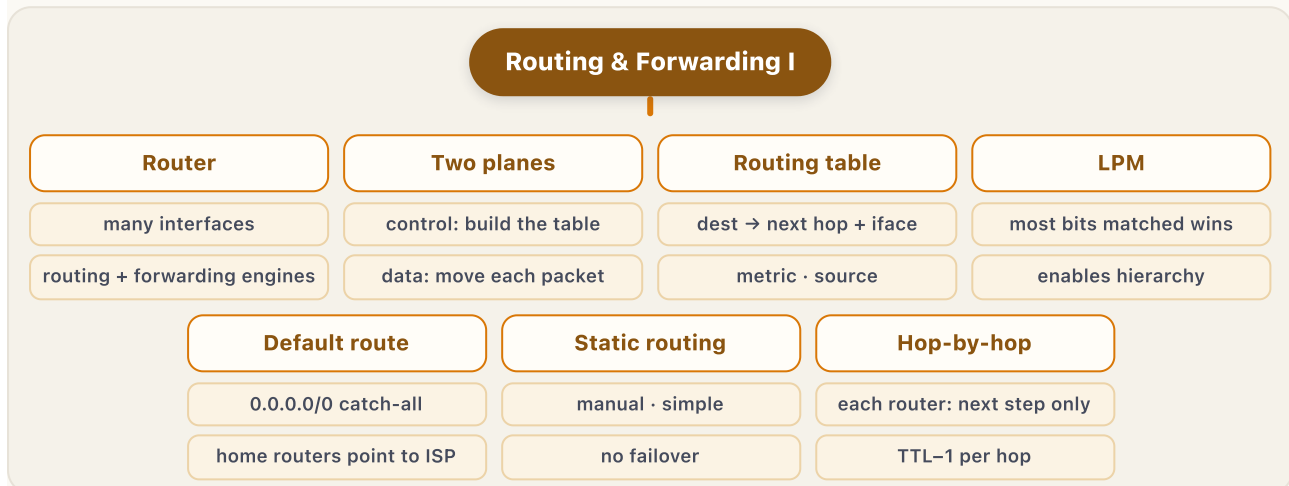
TTL

WHAT THIS LECTURE GIVES YOU

What a router actually does with your packet: the control plane vs data plane split, reading real routing tables, the Longest Prefix Match rule with worked lookups, static and default routes with Linux commands, a complete hop-by-hop forwarding trace, and the header fields that change at every hop — TTL, MACs, and why.

LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Distinguish **routing** (control plane) from **forwarding** (data plane)
- Read and interpret a **routing table** (destination, next hop, interface, metric)
- Apply **Longest Prefix Match** to pick the winning route
- Configure **static** and **default routes**; know when static fails
- Trace **hop-by-hop forwarding** and explain **TTL**



1 Routing vs Forwarding

Concept	Definition	When	Plane	Speed
Routing	Deciding WHERE packets should go — building/maintaining the routing table	Periodically / on topology change	Control plane	Slow, intelligent (algorithms, protocols)
Forwarding	Actually moving each packet out the correct interface	For every single packet	Data plane	Fast, mechanical (table lookup, ASIC hardware)

ANALOGY Routing = **planning the highway map** (surveying roads, updating the map). Forwarding = **reading a signpost at each junction** and turning. Drivers (packets) never see the map — only signposts.

What is a router?

- **Multiple network interfaces**, each attached to a different network
- A **routing table**: destination → next hop + interface
- A **routing engine** (control plane) that learns routes
- A **forwarding engine** (data plane) that processes packets at line speed

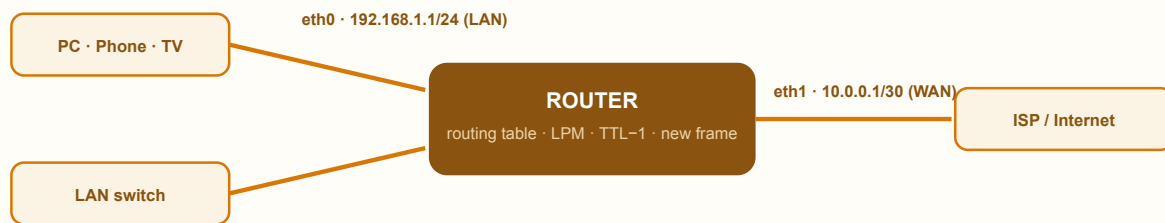


Figure 1. A router joins two or more networks — one interface per network.

EXAM

Router vs Switch: Switch = Layer 2, forwards by **MAC**, inside one LAN. Router = Layer 3, forwards by **IP + routing table**, **between** networks. Modern "home routers" bundle both (Lecture 11).

2 The Routing Table

Field	Meaning
Destination network	IP prefix this entry covers (e.g. 172.16.0.0/16)
Next hop	IP of the next router to hand the packet to ("directly connected" = deliver yourself)
Interface	Which port to send it out (eth0, eth1...)
Metric	Cost of this route — lower wins when several routes match equally
Source	How learned: connected · static · OSPF · BGP...

READING A REAL LINUX ROUTING TABLE

TERMINAL

```
ip route show
default via 192.168.1.1 dev eth0                # catch-all → gateway
10.0.0.0/30 dev eth1 proto kernel scope link src 10.0.0.1# directly connected
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.5
172.16.0.0/16 via 10.0.0.2 dev eth1 metric 100  # learned route
```

Example — Router R1 with three interfaces

Destination	Next hop	Interface	Source
192.168.10.0/24	directly connected	eth0 (Engineering LAN)	connected
192.168.20.0/24	directly connected	eth1 (Marketing LAN)	connected
10.1.1.0/30	directly connected	eth2 (WAN to ISP)	connected
0.0.0.0/0	10.1.1.2	eth2	static

3 Longest Prefix Match (LPM)

When a packet's destination matches **multiple entries**, the **most specific (longest) prefix wins**.

Worked lookup — destination 192.168.1.200

Table entry	Matches?	Bits matched	Result
192.168.0.0/16 via 10.0.0.1	✓ (192.168.x.x ⊆ /16)	16	
192.168.1.0/24 via 10.0.0.2	✓	24	
192.168.1.128/25 via 10.0.0.3	✓ (.200 is in .128–.255)	25	★ WINNER
0.0.0.0/0 via 10.0.0.4	✓ (matches everything)	0	

Decision: forward via **10.0.0.3** — 25 bits beat 24, 16 and 0.

WHY LPM MATTERS

It enables **hierarchical routing**: an ISP carries one route `10.0.0.0/8` for a million customers, while a customer with `10.45.0.0/16` gets the more specific (winning) route. Specific overrides general — that's also how BGP hijacks work (Lecture 8).

4 Default Route & Static Routing

Default route — 0.0.0.0/0

Matches **every** destination (zero bits checked).
The fallback when nothing else matches: "if in doubt, send here".

Who needs a default?

Home device → home router → ISP → Tier-2 → Tier-1. But a **Tier-1 core router has NO default** — it carries the full Internet table (900K+ routes) because there is nobody "above" it to default to.

STATIC ROUTING COMMANDS

TERMINAL

```
ip route add 192.168.20.0/24 via 10.0.0.1      # specific route
ip route add default via 10.0.0.1              # default route
ip route del 192.168.20.0/24 via 10.0.0.1     # remove
ip route show                                  # verify
# floating static (backup): high metric = used only if better route dies
ip route add 10.5.0.0/24 via 10.0.1.1 metric 200
```

Static routing

Assessment

Good for

Small networks · stub networks (one way in/out) · security-critical fixed paths · default gateways

Fails at scale

Manual → error-prone · no failover (dead next hop = black hole) · one entry per destination doesn't scale

5 Packet Forwarding — Hop by Hop

At EVERY router, for EVERY packet:

1. Extract destination IP
2. Search routing table — Longest Prefix Match
3. Match? → forward to next hop via that interface
4. No match & no default? → **DROP** + ICMP "Destination Unreachable"

Figure 2. The forwarding decision loop — millions of times per second.

Complete trace — Laptop 192.168.1.5 → Server 10.50.20.100

Laptop → R1 → R2 → R3 → Server

#	Where	Decision (LPM)	Frame MACs
1	Laptop	10.50.20.100 $\notin$ 192.168.1.0/24 → send to gateway 192.168.1.1	Laptop → R1
2	R1	10.50.0.0/16 → next hop 10.0.0.2 via eth1	R1 → R2 (<i>rebuilt</i>)
3	R2	10.50.0.0/16 → next hop 10.0.1.2 via eth1	R2 → R3 (<i>rebuilt</i>)
4	R3	10.50.20.0/24 directly connected → ARP for .100 → deliver	R3 → Server (<i>rebuilt</i>)
5	Server	Packet received — IP src/dst unchanged the whole way ✓	

KEY IDEA No router knows the full path. Each knows only the **next hop** — yet the chain of local decisions carries the packet across the planet. That's the "hop-by-hop" principle (and why routing loops are so dangerous — next lecture).

What changes at each hop?

Header field	Changes?	Why
IP Source / Destination	No (unless NAT)	End-to-end identity
TTL	Yes — -1 per router	Packet with TTL = 0 is dropped + ICMP Time Exceeded — prevents infinite loops
Ethernet Src/Dst MAC	Yes — rewritten	New frame per hop (L2 is local-only)

TTL & TRACEROUTE TTL starts at 64 (Linux/mac) or 128 (Windows). **traceroute exploits it:** send TTL=1 → router 1 drops & complains (revealing itself); TTL=2 → router 2 complains; ... until the destination answers. That's how Lecture 12 maps paths.

LECTURE 7 — RAPID REVISION CARD

Routing = build the table (control, slow); **forwarding** = per-packet lookup (data, fast).

Routing table: dest prefix → next hop + interface (+ metric, source).

LPM: longest matching prefix wins — /25 beats /24 beats /16 beats /0.

Default 0.0.0.0/0 matches everything; Tier-1 cores have none (full table).

Static: `ip route add NET via NEXT_HOP`; simple, no failover; floating = backup metric.

Forwarding loop: dest IP → LPM → next hop → new frame; no match → drop + ICMP Unreachable.

Per hop: TTL-1, MACs rewritten, IPs unchanged.

traceroute = increasing TTL + ICMP Time Exceeded replies.

✦ Important Questions — Exam Practice

A. Multiple Choice

Q1. Entries /8, /16, /24 and /0 all match destination 10.45.20.5. Which is used?

MCQ

- a) /0 — default is strongest
- b) /8
- c) /24 — most specific
- d) load-balanced across all

Q2. A packet's TTL reaches 0 at a router. The router:

MCQ

- a) forwards it anyway
- b) drops it and sends ICMP Time Exceeded
- c) returns it to source
- d) resets TTL to 64

Q3. Which command adds a default route via 10.0.0.1?

MCQ

- a) `ip route add 0.0.0.0/0 via 10.0.0.1`
- b) `ip default 10.0.0.1`
- c) `route default gateway 10.0.0.1`
- d) `ip gw 10.0.0.1`

Q4. A packet crosses 5 routers. How many times is the Ethernet frame rebuilt?

MCQ

- a) 0
- b) 1
- c) 5
- d) 4

Q5. Tier-1 core routers typically have:

MCQ

- a) many default routes
- b) no default route — the full Internet table
- c) only static routes
- d) no routing table

B. Short Answer (2–4 marks)

Q6. Differentiate routing and forwarding (plane, speed, frequency, intelligence).

SHORT

Q7. Explain Longest Prefix Match with the 192.168.1.200 example from this lecture.

SHORT

Q8. What is a default route and why doesn't a Tier-1 router have one?

SHORT

Q9. State three fields of a routing table entry and their meanings.

SHORT

Q10. Why does TTL exist? What does a router do when it hits zero, and which tool exploits this?

SHORT

C. Long Answer (5–8 marks)

Q11. Trace a packet from 192.168.1.5 to 10.50.20.100 through R1, R2, R3 (Figure/trace in §5): at each hop give the LPM decision, the new MAC pair, and the TTL change.

LONG

Q12. Compare static and dynamic routing: when is static appropriate? What are its two fatal weaknesses at scale, and what replaces it?

LONG

Q13. Routing table: 10.0.0.0/8 via A · 10.64.0.0/12 via B · 10.79.0.0/16 via C · 0.0.0.0/0 via D. For each destination — (i) 10.79.20.1 (ii) 10.80.1.1 (iii) 172.16.1.1 — state every matching entry and the chosen next hop.

NUMERICAL

Q14. "Hop-by-hop routing means no single device knows the complete path." Explain with the signpost analogy, and connect it to why routing loops must be prevented.

LONG

ANSWER KEY

A1. c — most specific wins (/24).

A2. b — drop + ICMP Time Exceeded to the source.

A3. a — or the equivalent "default via" form.

A4. c — every hop builds a fresh frame (5 routers → 5 rebuilds across the 5 links; IPs unchanged).

A5. b — nothing above them to default to; they hold ~900K+ specific routes.

A6. Routing: control plane, slow, periodic, algorithms. Forwarding: data plane, fast, per packet, table lookup.

A7. /25 (192.168.1.128/25) matches .200 with 25 bits → beats /24, /16 and default.

A8. 0.0.0.0/0 = "send everything unmatched here". Tier-1 = top of hierarchy; must know all prefixes explicitly.

A9. Destination prefix · next hop · interface (+ metric, source) — one line each.

A10. Prevents infinite looping packets; TTL=0 → drop + ICMP Time Exceeded; traceroute uses increasing TTLs.

A13. (i) 10.79.20.1 matches /8, /12 (10.64–79), /16 (10.79) and /0 → **C** (most specific). (ii) 10.80.1.1 matches /8 and /0 only (10.80 is outside 10.64–79) → **A**. (iii) 172.16.1.1 matches only /0 → **D**.

A11/A12/A14. Reproduce the §5 trace table and the signpost analogy; static's weaknesses = manual scale + no failover → dynamic protocols (L8).